

EXPERTSPEAK



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How do you use technology in your current setting?

This year the primary focus has been establishing the organization's internal processes. There has been a significant investment in cloud based data management systems that enable different staff teams to automate processes, reduce errors and give easy access to information across the country.

What challenges have you faced when trying to integrate technology in the classroom, especially since it's executive education?

The most common mistake that I have encountered when seeing technology being integrated into the classroom is that people forget that the technology is not the end solution in itself, but instead it is a tool that must be applied skilfully to enhance and support children otherwise it becomes an expensive experiment with limited or no impact on the children.

What positive outcomes has technology been responsible for?

I really enjoy watching technology being applied to help differentiate and adapt programs or content to an individual's needs. This can be through the accurate analysis of assessment data or through an adaptive online maths program that tailors questions to the child's responses. Additionally there is an immense power in enabling children to become independent learners through enhanced connectivity to online resources and communities which can stimulate a lifelong love of learning.

DID YOU KNOW?

- Students produce more data than any other demographic!
- Major publishers expect textbooks to be completely gone in 10 years!
- Only 54% of all kids in the world get access to a primary education, and only 22% get access to a secondary education; we are wasting almost 4/5ths of all human talent!
- In a "flipped classroom," students watch new instructional content at home, online, and discuss it in the classroom.
- In the past decade, VC investments in edtech have tripled from \$146 million in 2002 to \$429 million in 2011, according to the National Venture Capital Association Watch: <http://dgit.in/12QT0qR>

creative reasoning. We've seen a ton of development in the online learning space but really believe that highly-skilled teachers are essential in the development of a well-rounded student."

We list a few pathways that seem to hold a lot of promise in this regard. First, using technology, educators should be able to merge direct instruction and assessment into a seamless entity. A student would then be able to learn new content in small spurts while simultaneously demonstrating his/her mastery on each chunk before moving on a slightly more challenging objective (see box: "Masterclass: Knewton"). This should make post-lesson assessments redundant and thus reduce learning load on students and free up time for additional instruction.

Second, knowing that the pace at which a student learns may vary even in a small classroom, and also that it depends on the subject being taught, technology can be designed to help identify these minute differences, and direct teachers in meeting students where they presently stand.

Given the fact that it seems economically impossible to disseminate affordable technological solutions to all of our children in one go, a lot can still be achieved by using software and hardware tools to increase the quality of training that our teachers themselves get. A simple but highly effective way of doing this is by asking teachers to

record themselves in the classroom on their mobile phones, and seek feedback on them from their peers. Such micro-innovations are being promoted right here in India, by STiR. (More info: <http://www.stireducation.org/>)

As John Danner (Co-Founder of Rocketship Education) says, "In many places, there is no option to have an effective teacher in every small classroom of children. So technology is not a matter of replacement, but an acknowledgement that we are unlikely to have effective teachers in every classroom soon, so having the support structure that personalized learning creates will help us create more effective classrooms, provide more leverage (increased students which they can impact), and significantly increase pay for teachers. So the effects on teachers will be very positive, but will require people to rethink the role of teacher in a technology-rich environment vs. a traditional classroom."

Looking ahead over a 40 year horizon, the predominance of the classroom, as a framework for delivering education, is sure to reduce significantly. This will be replaced by "Studio" and "Virtual" setups. In a Studio structure, students will continue to meet in person, but utilize their time in peer-to-peer learning environments to discuss, learn and solve problems with each other, and the "teacher" will serve as a mere facilitator. This will be driven by technologies such as desk/wall-sized screens, online school communities, and flipped classrooms. The Virtual framework will allow participants to immerse themselves in high-quality learning experience irrespective of physicality and geography. Open courseware, book digitization, assessment algorithms, and algo-generated lessons will form the crux of the "virtual" educational experience.

As we leave Zeenat and her kids in their technologically aided journey of academic discovery (and hope that Peter Jackson does reply to her kids' emails, or at least incorporates their feedback in the sequel to *The Hobbit*), we cannot help but wonder how all the pieces of this puzzle may come together to provide high-quality education at affordable price points to all our children. **d**

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